Q-LINK Theoretical Extensions

Master Index and Hierarchical Ranking (13-Extension Study)

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Abstract

This document serves as the master index for the 13-part theoretical exegesis of the Q-LINK architecture. The study spans from initial topological ablations to the formulation of a universal bounding theorem. These extensions evaluate the limits of barren plateau mitigation via Dynamical Lie Algebra (DLA) expressibility and absolute gradient variance bounds. Below is the hierarchical rank-ordering of the explored branches (E1–E13, excluding E6), sorted in descending order of theoretical and practical importance.

Theoretical Hierarchy

The table below ranks the extensions from the mathematically optimal, generalized survivors down to the structural constraints and analytically pruned negative results.

Rank	ID	Theoretical Extension & Core Finding	Outcome
1	**E13**	**The Universal Bounding Theorem** Establishes the absolute limits of the ansatz regarding state purity, basis-blindness, and Hamiltonian symmetry fracturing.	Ultimate
2	**E12**	**The Fundamental Theorem of Coherent Residual Pathways** Distills the irreducible invariants (Dimensional, Trace, and Orthogonal) required for a coherent residual advantage.	Generalized
3	**E10**	**The Grand Unified Framework (R_{yy} + Periodic + m-Local)** Maximizes the combinatorial variance shield while providing independent spatial and temporal tuning.	Optimal
4	**E11**	**The Commutator Criterion & Directional Collapse** Defines the viable basis for alternative entanglers (CZ, CY) and proves the algebraic necessity of messenger-controlled directionality.	Prerequisite
5	**E7**	**The Period-Locality Manifold (Periodic + m-Local)** Proves temporal sparsity is orthogonal to the combinatorial trace shield, allowing independent window and floor tuning.	Essential
6	**E5**	**Combinatorial Cost-Function Scaling** Establishes that k-local cost functions actively amplify the Q-LINK structural trace advantage relative to Vanilla architectures.	Breakthrough
7	**E9**	**R_{yy} Local-Cost Isomorphism** Demonstrates that mapping to hardware-native YY-interactions preserves the spatial combinatorial advantage of local observables.	Viable
8	**E8**	**Periodic R_{yy} Temporal Manifold** Proves the algebraic isomorphism of the R_{yy} DLA is perfectly preserved under temporally sparse application.	Viable
9	**E4**	**Periodic Messenger (Temporal Sparsity)** Demonstrates that periodic application throttles operator spreading, delaying Haar saturation at finite depth.	Functional
10	**E2**	**Interaction Basis Ablation** Proves the structural requirement: R_{yy} preserves the $\mathfrak{su}(2^{n+1})$ expansion, while R_{zz} fatally collapses the DLA.	Constraint
11	**E3**	**Mechanism Ablation** A negative result proving the Q-LINK topology is irreducible; removing collection or distribution fractures the algebra.	Pruned
12	**E1**	**Multi-Messenger Scaling** A negative result proving that $k > 1$ messengers incur a catastrophic inverse-exponential Haar volume penalty.	Pruned